

For example, consider a 60-year-old female who invests her nest egg of \$100,000 among various equity mutual funds and wants to purchase a guarantee that would return the entire initial \$100,000 sum to her beneficiary upon her death. According to Table 9.1, she would need to pay 5.0 basis points (which is 0.05%) of her account value per year. Assume for a moment that the account grows to \$110,000 by the end of the first year and the guarantee fee is charged at the end of each year. This means that she would pay $0.05\% \times \$110,000 = \55 for this protection for the first year.

I don't want to get caught up in the mathematical details here, but these are model values, since it is very difficult to purchase capital market guarantees with maturities that are 40 to 50 years into the future. Yet, the mathematical models tell us that they are worth no more than a few basis points because they can theoretically be replicated for this cost. In plain English, just get some token life insurance. It will be cheaper.

Once again, an RoP guarantee essentially promises that at death you will get no less than your money back. That is the lower bound or worst-case scenario. There are some products and companies that offer a better worst-case scenario return, often called death benefits with look-back or ratchet guarantees, and those are worth a bit more, as illustrated in Table 9.2. The word “ratchet” refers to the fact that every time the account value within the variable annuity moves up (or ratchets) to a higher level, the death benefit is reset.

**TABLE 9.2 Capital Market Cost of a GMDB:
Look-Back (Ratchet) Guarantee**

Age	Female	Male
30	15.1 b.p.	25.0 b.p.
40	18.9 b.p.	31.6 b.p.
50	24.6 b.p.	41.8 b.p.
60	32.8 b.p.	56.4 b.p.
65	36.1 b.p.	62.5 b.p.

b.p. = basis point

Assumptions: Equity: expected returns = 6%, volatility = 20%

Source: Moshe Milevsky and the IFID Centre, 2008.